Warmup/Attendance: [Week 8: Friday 10/16/2026]

Full Name: SID:

Work with the people around you to try to solve these problems, which we'll discuss together once lecture begins. These are not graded for correctness, but you will turn this sheet in for attendance at the end of class.

1. 95% Confidence Interval for Sample Mean

```
## # A tibble: 1 × 3
##    xbar    sx     n
##   <dbl> <dbl> <int>
## 1  116.  78.4    36
```

What is an approximate 95% confidence interval for the mean air time in flights using the normal curve?

2. 95% Confidence Interval for Sample Proportion

```
## # A tibble: 1 × 2
##   p_hat     n
##   <dbl> <int>
## 1 0.143    49
```

What is an approximate 95% confidence interval for the proportion of United Airlines flights using the normal curve?

3. Increasing Precision

You would like to estimate a population mean using a simple random sample of n observations.

What sample size would you need to increase the precision of your estimate by approximately $3x$ compared to the original sample?

a) $n + 3$
b) $3n$
c) $n + 6$
d) $6n$
e) $9n$